HiTrace Solutions — LoanNBFC (Gidfine)

Phase 4 — Comparison API Fix & Convex MVP Architecture Proposal | Date: October 17, 2026 | Branch: development

1. Executive Summary & Integration Status

Phase 4 successfully resolved the contract gap identified during Phase 3 by implementing the missing Comparison API router (/api/comparison) mounted directly into the FastAPI application. The fix reuses existing schemas in api/schemas/comparison.py, multi-lender aggregation from mock_apis/lender_apis.py, dynamic weighting rules from utils/loan_ranker.py, and standard financial formulas from utils/financial_rules.py without fabricating any underwriting rules. The test suite executed with 6/6 tests passing, and the React 19 production build succeeded with zero errors.

Git Branch: development (Protected)	**Comparison Endpoints:** 4 Implemented & Tested	**Backend Test Suite:** 6/6 Passed (100%)
React 19 Build: Exit 0 (1,770 modules)	**TypeScript Errors:** 0 Errors	**Convex State:** Proposal Ready (Unconfigured)

2. Comparison API Trace & Schema Alignment

Frontend Hook / Operation	Backend Route & Method	Underlying Business Logic / Source	Contract Status
checkLenders()	GET /api/comparison/lenders	_load_lenders_data() (Ghana banks: Access, Fidelity, Zenith, etc.)	RESOLVED (200 OK)
getLoans()	POST /api/comparison/get-loans	aggregate_lender_offers() + LoanRanker weighted scoring	RESOLVED (200 OK)
selectLoan()	POST /api/comparison/select-loan	Lender verification, rate lookup & session binding	RESOLVED (200 / 404)
runWhatIf()	POST /api/comparison/what-if	Deterministic EMI delta & cost comparison via calculate_emi()	RESOLVED (200 OK)

3. Verification & Test Execution Results

• **Automated Test Suite (tests/test_comparison_api.py):** 6/6 tests executed and passed in 2.97s.
  - test_get_lenders: Verified 200 OK and schema fields across all 5 Ghanaian lenders.
  - test_get_loans_success: Verified 200 OK, dynamic weights, badges (■ BEST), and composite scores.
  - test_get_loans_validation_error: Verified 422 Unprocessable Entity on negative loan amount.
  - test_select_loan_success: Verified 200 OK for valid lender (access_bank_gh).
  - test_select_loan_not_found: Verified 404 Not Found on invalid lender ID.
  - test_what_if_simulation: Verified 200 OK with accurate EMI delta and total cost calculations.
• **Frontend Production Build:** npm run build (TypeScript 5.9 + Vite 5.4.21) completed in 35.25s with 0 errors.
• **Unexecuted Checks:** Real live database writes to remote MongoDB Atlas and live external credit bureaus were not called; mock mode was preserved.

4. Convex MVP Architecture Plan (React + Convex + FastAPI)

Architectural Separation of Concerns:
• **Convex (Real-Time Reactive Layer):** Replaces client-side polling with reactive WebSocket queries for UI reactivity.
  1. user_sessions: Active session token, current state, selected lender, UI view routing.
  2. chat_messages: Conversational thread between borrower and AI Loan Advisor with agent typing state.
  3. active_loans_cache: Read-only summary of disbursed loans, next EMI due date, and payment status.
  4. audit_notifications: User-facing toast events, document upload status, sanction readiness flags.
• **FastAPI / MongoDB (Authoritative Underwriting & Compliance Core):**
  - Rule-based underwriting engine, CIBIL bureau scoring, and DTI/FOIR compliance calculations.
  - Gemini Vision OCR document extraction & digital forgery forensics (never stored in Convex).

- Legally binding Sanction Letter PDF generation, E-Signature hashes, and Razorpay/disbursement execution.

• **Synchronization & Security Protocol:**
FastAPI publishes deterministic state updates to Convex via internal signed HTTP mutations (Convex Node SDK or webhook with shared secret). Borrower authentication uses JWTs verified by both Convex Auth and FastAPI middleware. Sensitive PII (national IDs, raw bank statements) remains strictly confined to MongoDB GridFS and is never duplicated into Convex.

5. Decisions Requiring Mathias's Approval

1. **Convex Provisioning:** Shall we initialize Convex (npx convex dev) and generate convex/schema.ts for the MVP?
2. **Vue Deprecation:** Now that React 19 is verified and comparison is functional, can we archive the 21 parallel .vue components?
3. **Comparison Session Storage:** Confirm if comparison sessions should persist in MongoDB/Convex or remain ephemeral in memory.